

ANSWER KEY

GATE CS — GATE CS 2026 Trend-Based Mock Paper
3

Q.No.	Type	Answer	Marks
1	3MCQ	—	3
2	2MCQ	—	2
3	3MCQ	—	3
4	2MCQ	—	2
5	2MCQ	—	2
6	2MCQ	—	2
7	2MCQ	—	2
8	2NAT	B	2
9	2MSQ	C	2
10	2MCQ	B	2
11	2MCQ	B	2
12	2NAT	B	2
13	2NAT	C	2
14	1MCQ	C	1
15	1MCQ	B	1
16	1MCQ	A	1
17	1MCQ	B	1
18	1MCQ	B	1
19	1NAT	C	1
20	1MCQ	B	1
21	1MCQ	C	1
22	2NAT	C	2
23	1NAT	C	1
24	1MCQ	B	1
25	1NAT	A	1
26	2MCQ	C	2
27	1MCQ	A	1
28	1MCQ	C	1
29	1MCQ	B	1
30	1MCQ	B	1
31	1MCQ	B	1
32	1NAT	B	1
33	1MCQ	C	1
34	1NAT	C	1
35	1NAT	B	1
36	1MCQ	B	1

37	1MCQ	B	1
38	1NAT	A	1
39	1MCQ	B	1
40	1MCQ	B	1
41	1NAT	B	1
42	1MCQ	B	1
43	1MCQ	B	1
44	1MCQ	C	1
45	1MCQ	A	1
46	2NAT	—	2
47	2MCQ	—	2
48	2MSQ	—	2
49	2NAT	—	2
50	2MCQ	—	2
51	2MCQ	—	2
52	2MCQ	—	2
53	2MCQ	—	2
54	2MCQ	—	2
55	2MSQ	—	2
56	1MCQ	B	1
57	2MCQ	—	2
58	2NAT	—	2
59	2MCQ	—	2
60	2NAT	—	2
61	2MCQ	—	2
62	2NAT	—	2
63	1MCQ	—	1
64	2MSQ	—	2
65	2MSQ	—	2

Q1. General Aptitude | 3 marks | 3MCQ

According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is re...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q5

Q2. General Aptitude | 2 marks | 2MCQ

If ' ' denotes increasing order of intensity, then the meaning of the words [walk jog sprint] is analogous to [bothered _...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q1

Q3. General Aptitude | 3 marks | 3MCQ

In the sequence 6, 9, 14, x, 30, 41, a possible value of is

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q5

Q4. General Aptitude | 2 marks | 2MCQ

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of s...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q3

Q5. General Aptitude | 2 marks | 2MCQ

Two wizards try to create a spell using all the four elements, water, air, fire, and earth. For this, they decide to mix...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q2

Q6. General Aptitude | 2 marks | 2MCQ

Consider the relationships among P, Q, R, S, and T: P is the brother of Q. S is the daughter of Q. T is the sister..

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q4

Q7. General Aptitude | 2 marks | 2MCQ

The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q4

Q8. General Aptitude | 2 marks | 2NAT

Antonym of 'ephemeral':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q10

Q9. Engineering Mathematics | 2 marks | 2MSQ

Probability of at least one head in 3 coin tosses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q11

Q10. Engineering Mathematics | 2 marks | 2MCQ
Derivative of x^3 at $x=2$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q1

Q11. Engineering Mathematics | 2 marks | 2MCQ
Determinant of 2×2 identity:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q1

Q12. Engineering Mathematics | 2 marks | 2NAT
Mean of binomial $n=10, p=0.5$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q1

Q13. Engineering Mathematics | 2 marks | 2NAT
Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q1

Q14. Programming and Data Structures | 1 marks | 1MCQ
Avg hash lookup:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q3

Q15. Programming and Data Structures | 1 marks | 1MCQ
Infix to postfix uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q6

Q16. Programming and Data Structures | 1 marks | 1MCQ
Inorder traversal of BST gives:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q3

Q17. Programming and Data Structures | 1 marks | 1MCQ
What is the time complexity of inserting at the beginning of a linked list?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q3

Q18. Algorithms | 1 marks | 1MCQ
BFS uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q7

Q19. Algorithms | 1 marks | 1NAT
Merge sort space:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q7

Q20. Algorithms | 1 marks | 1MCQ
Worst-case time complexity of merge sort:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q12

Q21. Algorithms | 1 marks | 1MCQ
LCS uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 CS Session 1 Q7

Q22. Algorithms | 2 marks | 2NAT
The time complexity of Dijkstra's algorithm using a binary heap is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q7

Q23. Algorithms | 1 marks | 1NAT
Kruskal uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q2

Q24. Algorithms | 1 marks | 1MCQ
Time complexity of binary search:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q11

Q25. Operating Systems | 1 marks | 1NAT
TLB stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 CS Session 1 Q9

Q26. Operating Systems | 2 marks | 2MCQ
Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2...

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q8

Q27. Operating Systems | 1 marks | 1MCQ
Round-robin uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q4

Q28. Operating Systems | 1 marks | 1MCQ
Which of the following is NOT a necessary condition for deadlock?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q4

Q29. Operating Systems | 1 marks | 1MCQ
Inode contains:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q4

Q30. Operating Systems | 1 marks | 1MCQ
Semaphore with value 1 is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q8

Q31. Operating Systems | 1 marks | 1MCQ
Banker's algorithm is for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q4

Q32. Computer Networks | 1 marks | 1NAT
OSPF is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q6

Q33. Computer Networks | 1 marks | 1MCQ
Asymmetric encryption:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q5

Q34. Computer Networks | 1 marks | 1NAT
Which protocol operates at the application layer of the TCP/IP model?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q6

Q35. Computer Networks | 1 marks | 1NAT
CSMA/CD is used in:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q13

Q36. Computer Networks | 1 marks | 1MCQ
TCP connection establishment uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q14

Q37. Computer Networks | 1 marks | 1MCQ
Protocol to find MAC from IP:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q6

Q38. Digital Logic | 1 marks | 1NAT
Flip-flop stores:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 CS Session 1 Q2

Q39. Digital Logic | 1 marks | 1MCQ
Full adder has how many inputs:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q2

Q40. Digital Logic | 1 marks | 1MCQ
The number of Boolean functions of n variables is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q2

Q41. Theory of Computation | 1 marks | 1NAT
Which language is regular?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q13

Q42. Theory of Computation | 1 marks | 1MCQ
Halting problem:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q8

Q43. Theory of Computation | 1 marks | 1MCQ
PDA recognizes:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q14

Q44. Compiler Design | 1 marks | 1MCQ
3-address code operands:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q9

Q45. Compiler Design | 1 marks | 1MCQ
Lexical analyzer outputs:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q15

Q46. CS | 2 marks | 2NAT
Consider routing table of an organization's router shown below: Subnet Number Subnet Mask Next Hop 12.20.164.0 255.255.2...

Answer: See explanation

Source: Adapted from GATE 2022 CS Session 1 Q55

Q47. CS | 2 marks | 2MCQ
A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stall...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 1 Q56

Q48. CS | 2 marks | 2MSQ
Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size),...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q54

Q49. CS | 2 marks | 2NAT
Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mod...

Answer: See explanation

Source: Adapted from GATE 2023 CS Session 1 Q23

Q50. CS | 2 marks | 2MCQ
In the given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q9

Q51. CS | 2 marks | 2MCQ

Consider a sequence a of elements $a_0 = 1, a_1 = 5, a_2 = 7, a_3 = 8, a_4 = 9$, and $a_5 = 2$. The following operations are performed...

Answer: See explanation

Source: Adapted from GATE 2023 CS Session 1 Q59

Q52. CS | 2 marks | 2MCQ

Consider the following C program: `#include <stdio.h> void fX(){ void fX(); char a; int main(){ if((a=getchar()) != '\n')...`

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 1 Q19

Q53. CS | 2 marks | 2MCQ

Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B ...

Answer: See explanation

Source: Adapted from GATE 2023 CS Session 1 Q53

Q54. CS | 2 marks | 2MCQ

Let $f : A \rightarrow B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation on the ...

Answer: See explanation

Source: Adapted from GATE 2023 CS Session 1 Q49

Q55. CS | 2 marks | 2MSQ

Consider the following definition of a lexical token id for an identifier in a programming language, using extended re...

Answer: See explanation

Source: Adapted from GATE 2023 CS Session 1 Q19

Q56. Digital Logic | 1 marks | 1MCQ

The number of Boolean functions of n variables is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q2

Q57. CS | 2 marks | 2MCQ

The symbol $\twoheadrightarrow$ indicates functional dependency in the context of a relational database. Which of the following options is/are...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 1 Q44

Q58. CS | 2 marks | 2NAT

Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data element z by a transaction T_i , respectively. Consider ...

Answer: See explanation

Source: Adapted from GATE 2022 CS Session 1 Q39

Q59. CS | 2 marks | 2MCQ

Consider the following statements: (i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding MAC address...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q16

Q60. CS | 2 marks | 2NAT

Consider the following C program: `#include <stdio.h> void fX(){ void fX(); char a; int main(){ if((a=getchar()) != '\n')...`

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 1 Q19

Q61. CS | 2 marks | 2MCQ

The number of distinct minimum-weight spanning trees of the following graph is _____ b 3 c 1 2 2 1 a d 2 g 2 1 2 1 2...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q59

Q62. CS | 2 marks | 2NAT

Let $(\rightarrow, \rightarrow, \rightarrow)$ be a context-free grammar in Chomsky Normal Form with $\rightarrow = \{ \rightarrow, \rightarrow \}$ and containing 10 variable symbols including...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 1 Q59

Q63. General Aptitude | 1 marks | 1MCQ

If $\rightarrow = -$ for all real values of $\rightarrow$, which one of the following statements is true?

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q3

Q64. CS | 2 marks | 2MSQ

Let $\rightarrow$ be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q51

Q65. CS | 2 marks | 2MSQ

A meld operation on two instances of a data structure combines them into one single instance of the same data structure....

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q38